Topic 2. Concepts of Probability and Probability Rues

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Contents

Q: Why do we need to study probability? **A**: We can build a bridge between a sample and its population so that one can make inferences about the population from its samples.

1 Introduction

In this note, we introduce the definitions of probability, random variables, and problems associated with a distribution based on a special distribution: uniform distribution.

2 Definitions of Probability

We need to use the following concepts to define or approximate the probability of an event.

2.1 Experiment and Event

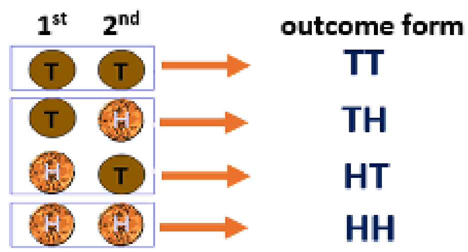
An experiment is any process that generates well-defined outcomes. On any single repetition of an experiment, one and only one of the possible experimental outcomes will occur.

Example 1. The following table gives several examples of experiments.

Experiments	***Experimental Outcomes***
Toss a coin	Heads, Tails
Select a part for inspection	Defective, non-defective
Conduct a sales call	Purchase, no purchase
Roll a dice (six-sided)	1, 2, 3, 4, 5, 6
Play a football game	win, lose, tie

The Sample Space of an experiment is the set of all possible outcomes. We usually use a capital letter in Greek or English to denote the sample space. For example, Ω, S, or U.

Example 2. Consider the experiment of tossing a balanced coin sequentially for 2 times. We use H to denote the heads and T to denote the tails.



Therefore, the sample space of this experiment is $\Omega = \{TT, TH, HT, HH\}$.

An Event is a subset of outcomes from the sample space. We usually use a capital letter to denote an event.

Example 3. We define Event E to be at least one H is observed in the coin tossing example.

Solution: The outcomes in the subset $\{HT, TH, HH\}$ satisfy the requirement in the definition. Therefore, event E is defined by $E = \{HT, TH, HH\}$.

3 Definition of Probabilities

Notations: We use P or Pr to denote the probability and $A, B, C, \cdot$, to denote specific events. $P(A)$ is the probability that event A occurs.

3.1 Definitions of Probability

- **Method 1:** Relative Frequency Approximation of Probability.

Conduct or observe an experiment a large number of times and count the number of times that event A occurs. Based on these actual results, P(A) is estimated as follows:

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P(A)= (# of times of A being observed) / (# of repeated trials)
```

- **Method 2:** Classical Approach to Probability (Requires Equally Likely Outcomes of the experiment)

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P(A) = (# of outcomes in A) / (# of outcome in S)
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3.2 Some Examples

Example 4. **Guessing Answers on an ACT:** A typical multiple-choice question has 5 possible answers. If you make a random guess on one question, what is the probability that your response is wrong?

Solution: Method 2 classical approach applies to this question since each of the choices is random, that is, each of the 5 letters A, B, C, D, and E are equally likely to be selected as the correct answer.

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Pr(wrong answer) = 4 / 5 =0.8
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Example 5. **Gender of Children**: Find the probability that a randomly selected couple with 3 children will have exactly 2 boys. Assume that boys and girls are equally likely, and the gender of any child is not influenced by the gender of any other child.

Solution Based on the given condition, this is an equally likely experiment. We first use a table to list all possible outcomes in the sample space. Let B = boy and G = girl. Then the following table lists all possible outcomes of this experiment.

1st	2nd	3rd		Outcome form
B	**B**	**B**	→	**BBB**
B	**B**	**G**	→	**BBG**
B	**G**	**B**	→	**BGB**
G	**B**	**B**	→	**GBB**
G	**G**	**B**	→	**GGB**
G	**B**	**G**	→	**GBG**
B	**G**	**G**	→	**BGG**
G	**G**	**G**	→	**GGG**

We list the sample space as follows

```
S = { BBB   BBG   BGB   GBB   GGB   GBG   BGG   GGG }
E =  having exactly 2 boys = { BBG   BGB   GBB }.
```

The desired probability can be calculated as

```
Pr(E)= #E/#S  = 3 / 8 =0.375.
```

3.3 Discrete Probabilities on Contingency Tables

Contingency tables classify outcomes in rows and columns. Table cells at the intersections of rows and columns indicate frequencies of both events coinciding.

Example 6. The following table displays events for computer sales at a fictional store.

	PC	**Mac**	**Row Totals**
Male	66	40	106
Female	30	87	117
Column Totals	96	127	223

Specifically, it describes the frequencies of sales by the customer's gender and the type of computer purchased. The cells' counts represent the number of PCs and Macs purchased by both genders.

Finding the following probabilities:

(1). Randomly select a female customer, what is the probability that she bought a Mac computer?

(2). Randomly select a customer from those who bought a computer from the store, what is the probability that the customer is a male?

(3). Randomly select a customer who bought a computer from the store, what is the probability she/he bought a Windows computer?

(4). Randomly select a customer who bought a computer from the store, what is the probability that the customer is female and bought a Mac computer?

Solution (1). $P(\text{Mac among female}) = 87/117 = 0.744$

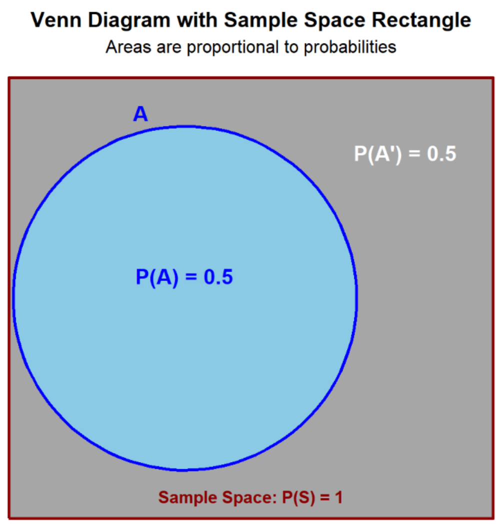
(2). $P(\text{male among customers who bought a computer}) = 106/223 = 0.4753$

(3). $P(\text{windows among all computers sold}) = 96/223 = 0.4305$

(4). $P(\text{female \& bought a mac computer}) = 87/223 = 0.3901$

4 Venn Diagram Representation

A Venn diagram shows the sample space, S, as a unit area (typically a rectangle), and the probabilities of events as areas whose sizes are the probabilities of the events. The size of overlap between areas indicates the probability of intersections of events.

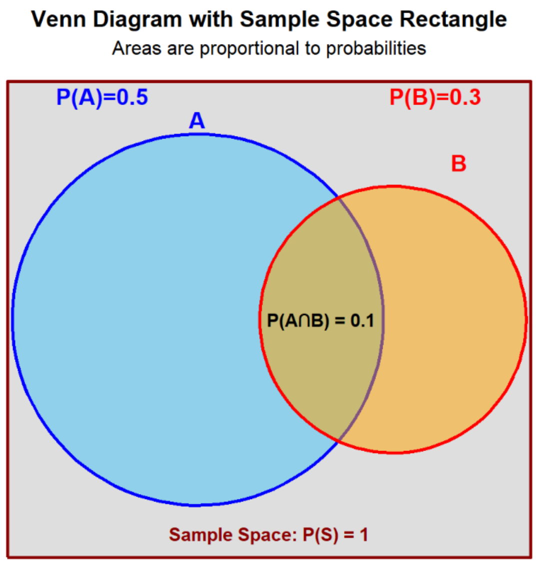


Sample Space (S): The **entire rectangle** (including the circular region) represents all possible outcomes of an experiment.

Event A: Circles (**sky blue region**) within the rectangle represent a specific event or subset of the sample space.

Complement (A'): Elements not in A (**dark gray region**).

Multiple Events Representation

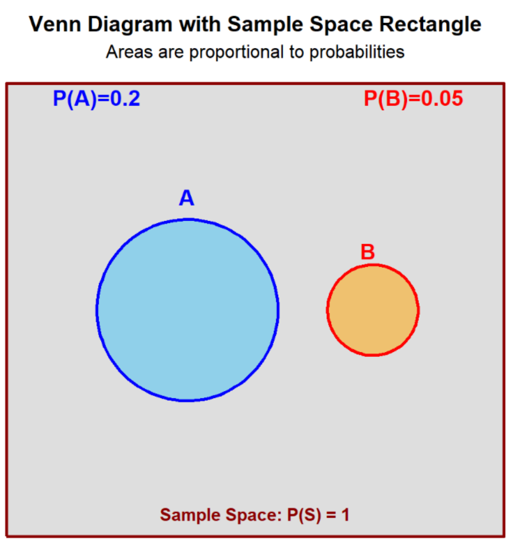


Key Regions defined based on existing events **A** and **B** in Venn Diagrams:

Union (A ∪ B): All elements in **either A or B or both**.

Intersection (A ∩ B): Elements common to **both A and B** (see the above figure).

Mutually Exclusive: No overlap between events. The following Venn diagram shows the exclusive events **C** and **D**.

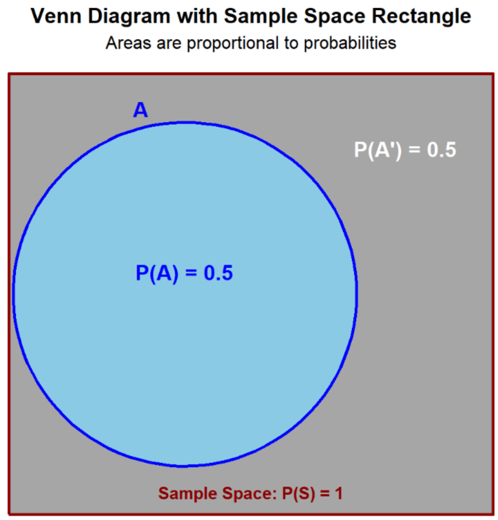


5 Basic Probability Rules

Probability measures an event's likelihood from 0 (impossible) to 1 (certain). We have introduced the three operations to define **new events** based on the existing events. The goal is how to find the probability of the derived **events** based on the probabilities of existing events. This section output the logic and rules of finding the probability of the derived **new events** using the following basic rule.

5.1 The Complement Rule

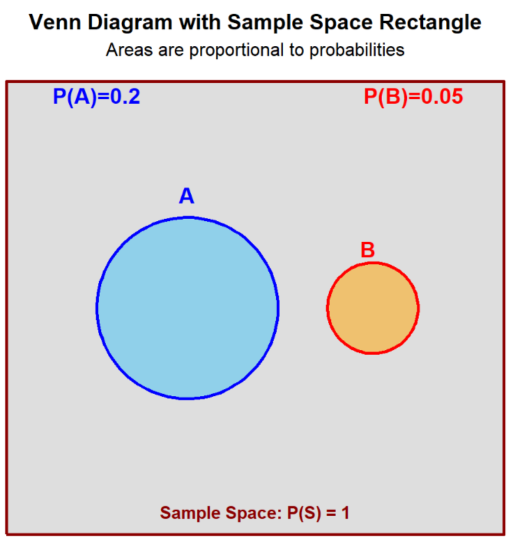
The complement rule states that the probability an event does not happen is 1 minus the probability it does. According to the definition, the rule is depicted in the following figure (which is also given previously)



5.2 Additition Rule

The addition rule (also called additive rule) finds the probability of either event A or B occurring;

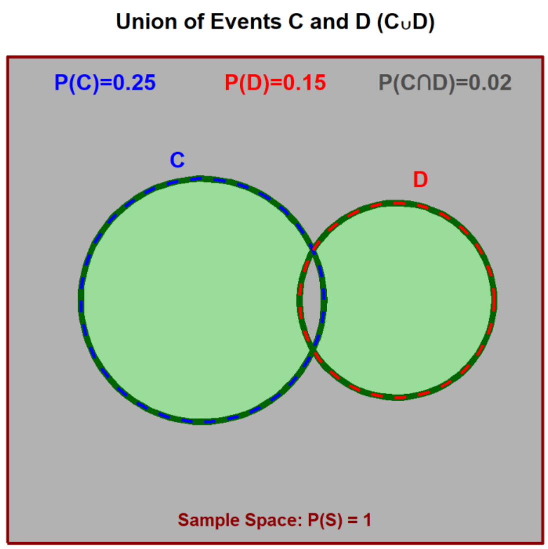
- For mutually exclusive events (the two events are not overlaped!), we add their probabilities.



That is,

$$P(A \cup B) = P(A) + P(B).$$

- For overlapping events, we must subtract the probability that both occur to avoid double-counting (see the following figure).



That is,

$$P(C \cup D) = P(C) + P(D) - P(C \cap D).$$

Example 1. A coffee shop finds that on a Monday morning:

60% of customers order coffee.

40% of customers order a pastry.

25% of customers order both coffee and a pastry.

If a random customer is selected, what is the probability that they:

a) Order coffee or a pastry (or both)?

b) Order neither coffee nor a pastry?

Solution:

Let C = event that a customer orders coffee. Then $P(C) = 0.60$

Let P = event that a customer orders a pastry.Them $(P) = 0.40$ and $P(C \cap P) = 0.25$

(a) This is a classic "OR" problem. The events are not mutually exclusive (a customer can order both), so we use the general addition rule:

$$P(C \cup P) = P(C) + P(P) - P(C \cap P) = 0.60 + 0.40 - 0.25 = 0.75.$$

Conclusion: The probability a customer orders coffee, a pastry, or both is 75%.

(b). "Ordering neither" is the complement of "ordering coffee or a pastry." We use the complement rule:

$$P(\text{Neither}) = 1 - P(C \cup P) = 1 - 0.75 = 0.25$$

Therefore, the probability a customer orders nothing is 25%.

5.3 Conditional Probability and Multiplication Rule

Conditional probability defines the probability of an event occurring based on a given condition or prior knowledge of another event.

It is the **likelihood** of an event occurring, given that another event has already occurred (i.e., this is prior information). In probability, this is denoted as **A** given **B**, expressed as **P(A | B)**, indicating the probability of event A when the event B has already occurred.

We define two events **Rain** and **Umbrella** based on the following figure.

Based on the definition of probability and rules, we have

- P(**Rain**) = 3/10 = 0.3.
- P(**Umbrella**) = 5/10 = 0.5.
- P(**Rain** ∩ **Umbrella**) = 2/10 = 0.2.
- P(**Rain** ∪ **Umbrella**) = 6/10 = 0.6.

We want to find the probability of using an umbrella given the prior information that it rains; this is the conditional probability **P(Umbrella | Rain)**. Intuitively, this probability is only concerned with the days when it rains. That is, it represents the proportion of rainy days on which an umbrella is used.

$$P(\text{Umbrella}|\text{Rain}) = \frac{\text{number of(Umbrella} \cap \text{Rain)}}{\text{Number of Total Rainny Days}} = \frac{2}{3}.$$

The reasoning and answer provided above are correct. **It is important to note that the sample space for this conditional probability is restricted to rainy days, not the entire 10 days. Unconditional probabilities, however, are defined based on the complete sample space of all 10 days.** To keep consistency of using the same sample space to define all related probability, we can modify the above conditional probability formula in the following

$$P(\text{Umbrella}|\text{Rain}) = \frac{\text{number of (Umbrella} \cap \text{Rain)/10}}{\text{Number of Total Rainny Days/10}} = \frac{2/10}{3/10} = \frac{2}{3}.$$

The formal definition of the conditional probability uses the same logic and is defined by

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

The core idea of conditional probability is to use the information that a prior event (B) has occurred to improve the estimate of the probability of the future event (A). The numerator $P(A \cap B)$ represents the probability both A and B occur and denominator $P(B)$ is the probability of the occurrence of event B.

Independent Events: Two events are **independent** if $P(A|B) = P(A)$. That is, the information of B does not impact the probability of A.

Properties: If two events A and B are independent, their complements A' and B' are also independent.

The multiplication rule finds the probability both A and B occur;

- For independent events, we multiply their probabilities, while
- For dependent events, we multiply the probability of the first by the conditional probability of the second given the first.

From the definition of the conditional probability, we have

$$P(A \cap B) = P(B) \times P(A|B).$$

By combining the definitions of conditional probability and independent events, we can derive the following rule to test whether two events are independent.

$$P(A \cap B) = P(A) \times P(B).$$

Example 2: A factory has two independent production lines, Line A and Line B.

The probability that Line A produces a defective item is 2%.

The probability that Line B produces a defective item is 3%.

An inspector randomly selects one item from Line A and one from Line B. What is the probability that:

a) Both items are defective?

b) Neither item is defective?

c) At least one item is defective?

Solution:

Because the lines are **independent**, what happens on one line does not affect the other.

Let D_A = item from Line A is defective. Then $P(D_A) = 0.02$

Let D_B = item from Line B is defective. Then $P(D_B) = 0.03$

(a). **Both are defective** is an **AND** problem for independent events. Using the multiplication rule of independent events, we have

$$P(\text{Both Defective}) = P(D_A \cap D_B) = P(D_A) \times P(D_B) = 0.02\text{Ö}0.03 = 0.0006.$$

The event **Neither is defective** is an **AND** problem. The word **ether** implies a negation, meaning we are dealing with the intersection of the complements of the individual **defective** events.

$$P(\text{ Not } D_A) = P(D'_A) = 1 - 0.02 = 0.98.$$

$$P(\text{ Not } D_B) = P(D'_B) = 1 - 0.03 = 0.97.$$

The property of **independent** events indicates that the two complementary events are also indepedent, which implies that

$$P(\text{Neither Defective}) = P(\text{ Not } D_A \cap \text{ Not } D_B) = 0.98Ö0.97 = 0.9506.$$

Hence, there is a 95.06% chance that both items are fine.

(c). **At least one is defective** is the complement of **neither is defective.**

$$P(\text{ At least one defective}) = 1 - P(\text{Neither Defective}) = 1 - 0.9506 = 0.0494.$$

Therefore, there is a 4.94% chance that at least one of the two items is defective.

Example 3: A company classifies its employees by gender and whether they have a university degree.

Has Degree	No Degree	Total
Male	8	12
Female	10	5
Total	18	17

If one employee is selected at random, what is the probability:

a) The employee is female, given they have a degree?

b) The employee has a degree, given they are female?

Solution: This problem shows how conditional probability can change the sample space.

(a): In $P(\text{Female}|\text{ Degree})$, the condition is **has a degree.** So, we only look at the **Has Degree** column.

- Total with a degree: 18
- Females with a degree: 10

$$P(\text{ Female } | \text{ Degree}) = \frac{10}{18} = \frac{5}{9}.$$

That is, given an employee has a degree, there is a 5/9 chance they are female.

(b). In $P(\text{Degree } | \text{ Female})$, the condition is **is female**. So, we only look at the **Female** row.

- Total females: 15
- Females with a degree: 10

$$P(\text{Degree } | \text{ Female}) = \frac{10}{15} = \frac{2}{3}.$$

Consequently, given an employee is female, there is a 2/3 chance they have a degree.

5.4 Summary of Probability Rules

The above statements are the basic probability rules. We summarize them in the following table.

Rule	**Scenario**	**Formula**
Complement	Event A does **not** happen	$P(A') = 1 - P(A)$
Addition (OR)	Mutually Exclusive Events	$P(A \cup B) = P(A) + P(B)$
Addition (OR)	Non-Mutually Exclusive Events	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Multiplication (AND)	Independent Events	$P(A \cap B) = P(A) \times P(B)$
Multiplication (AND)	Dependent Events	$P(A \cap B) = P(A) \times P(B\|A)$
Conditional	Probability of B given A	$P(B\|A) = \dfrac{P(A \cap B)}{P(A)}$

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6 Basics of Random Variables